

CHronoZ

MACRO INTELLIGENCE AGENT

Empirical Economic Benchmark, State, Pricing & Decision Architecture

A clearer quantitative whitepaper for turning real economic data, new research dumps, market pricing and future outcomes into a measurable, testable and compounding macro research system.

CANONICAL BASELINE - v0.2

This version replaces the hard-coded economic-state concept in v0.1 with an empirical benchmark and state engine. Dump-to-Text remains the canonical input gateway. Python/Jupyter becomes the quantitative research layer; LLM agents remain the interpretation and research layer. The document is a system specification, not investment advice.

QuantFun x Wonnyy / 16 September 2026

WHITEPAPER v0.2

Abstract

CHronoZ is a temporal macro intelligence system designed to turn fragmented research into an auditable economic worldview that can be measured, challenged, backtested and improved through time. It accepts raw research dumps - PDFs, screenshots, charts, tables, notes, speeches, research reports and Markdown - converts them into structured evidence, compares that evidence with an empirical economic state, investigates causal hypotheses, evaluates expectations and market pricing, and finally produces a dated Macro Journal and machine-readable research state.

The core change in v0.2 is simple but fundamental: CHronoZ should not begin from a manually hard-coded causal graph or an LLM-generated idea of what is "normal." The economic baseline must first be estimated from real historical data. Current conditions are then measured as deviations from that benchmark. LLMs may interpret those measurements, but they may not invent them.

What changed from v0.1

v0.1 idea	v0.2 replacement
Current Default Economic State created manually	Empirical Economic Benchmark + Current Economic State calculated from data
Transmission graph constrains causal reasoning	Transmission map becomes a library of candidate mechanisms; causality is a hypothesis to investigate
TypeScript-first economic logic	Python/Jupyter is first-class for statistics, time series, ML and calibration; TypeScript remains the application/agent layer
State may be updated by qualitative evidence	Text evidence does not directly change numeric state; it supports, contradicts or expands a thesis
Financial conditions mixed into the economy	Real economic state and market-pricing state are separated to reduce circularity
Regime mainly a narrative compression	Regime can be discovered statistically from state history, then interpreted by LLM/human
Historical memory as journals	Journals remain, but each timestamp also stores state, evidence, pricing, outlook and later realized outcomes

Plain-language summary

CHronoZ has two brains. The quantitative brain measures the economy. The LLM brain reads, researches and explains what those measurements may mean. Neither should pretend to do the other's job.

Contents

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1. Product Thesis

Macro research does not arrive as one clean dataset. Useful information comes as releases, charts, screenshots, bank research, speeches, notes, market moves, policy announcements and qualitative observations. The problem is therefore not just retrieving data. It is deciding what is measurable, what is merely suggestive, what matters to the current state, what is already priced, and what eventually proved right or wrong.

The weak pattern

Raw input -> LLM -> confident narrative

Fast, but difficult to audit. The model may blur facts, assumptions, historical norms, causal stories and market implications into one fluent answer.

The CHronoZ pattern

Raw Dump -> Text/Evidence -> Benchmark/State -> Research -> Pricing -> Outlook -> Decision -> Outcome -> Calibration

Each stage has a different job and a different standard of evidence.

Founder design principles

- Dump-to-Text remains the first gateway. CHronoZ should be able to receive messy human research material before any thesis is formed.
- The economic baseline is empirical. "Neutral" is estimated from historical data or a clearly documented economic anchor - not invented by an LLM.
- Current economic state is a vector, not one magic score. Weak growth and high inflation must not cancel each other into a fake neutral reading.
- Numeric state is calculated by code. LLMs may explain numbers; they may not manufacture them.
- New text evidence does not directly rewrite the numeric state. It is first mapped, compared, researched and tested.
- Causality is a hypothesis to investigate. Candidate transmission mechanisms can guide research, but they are not hard-coded economic truth.
- Economic reality, consensus expectations and market pricing are separate objects.
- Regime labels are summaries. The underlying state remains visible and auditable.
- Backtests must be walk-forward and must respect data vintages and release timing.
- The system should learn from its own history: what was known, what was believed, what was priced, what happened next, and where the errors came from.

2. System Architecture

The v0.2 architecture is easier to reason about if CHronoZ is separated into four layers. The first sees the material. The second measures the economy. The third reasons about meaning and causality. The fourth stores history so the system can learn through time.

Layer	Primary job	Main technology
1. Perception	Turn PDF/image/chart/note/report into canonical text + structured evidence	Multimodal model / parser / extraction code
2. Measurement	Build benchmark, state, momentum, regimes and market measurements	Python, statistics, time series, ML
3. Reasoning	Research gaps, causal hypotheses, expectations, interpretation, critique, journal	LLM agents + tools
4. Memory	Store state, evidence, thesis, provenance and future outcomes	Librarian / Wonnyy / JSON + Markdown

Canonical flow

RAW DUMP -> DUMP-TO-TEXT -> STRUCTURED EVIDENCE -> ECONOMIC STATE -> CAUSAL RESEARCH -> PRICING -> OUTLOOK -> DECISION -> OUTCOME

The state engine is loaded before new evidence is interpreted.

Two brains, one system

Quantitative brain	Reasoning brain
Historical data	Research documents
Transforms / normalization	Semantic classification
Benchmark distribution	Information-gap detection
State vectors	Causal hypotheses
Momentum / breadth	Contradictions / alternatives
Regime models	Narrative synthesis
Pricing measurements	Cross-asset interpretation
Backtests / calibration	Critic / falsification

Hard boundary

LLM may describe a numeric state, but the numeric state must come from the quantitative engine. Python may detect correlation or regime structure, but it does not automatically prove economic causality.

3. Dump-to-Text & Evidence Layer

CHronoZ begins with a dump, not with a clean database. The user may provide a PDF, screenshot, chart, table, note, speech, research report or Markdown file. Dump-to-Text converts that heterogeneous material into a canonical representation that downstream systems can work with.

Dump-to-Text is more than OCR

- Extract readable text while preserving page/section provenance.
- Identify dates, entities, countries, markets, policy institutions and named indicators.
- Extract numbers and units when present.
- Describe charts and tables instead of flattening them into meaningless text.
- Preserve the original source reference so every later claim can be traced back.
- Flag uncertainty when the image, chart or wording is ambiguous.

Canonical evidence object

Example

Source: bank research PDF
Date: 2026-09-16
Geography: US
Claim: "Banks tightened lending standards."
Candidate node: Credit
Direction: Tightening
Horizon: Current / forward-looking
Provenance: page 7, lending survey section

Cold evidence rule

The extraction layer should be intentionally boring. It should not be rewarded for sounding insightful. Its job is to say what the material contains, not what CHronoZ wants it to mean.

4. Economic Benchmark Engine

The benchmark answers one question: "What has historically counted as normal for this transformed economic variable or economic node, using only information that would have been available at that time?" It is not a forecast and it is not a market fair-value model.

4.1 Benchmark is not one average

A useful benchmark contains at least three pieces: a center, a measure of dispersion, and the historical distribution. The black line in the founder sketch is only the center of the benchmark; the blue bands represent dispersion around it.

Benchmark_{i,t} = { center_{i,t} , dispersion_{i,t} , historical distribution_{i,t} }

The exact estimator can differ by variable.

4.2 Why raw averaging is wrong

GDP growth, unemployment, CPI, PMI and industrial-production indices use different units and scales. Adding their raw levels is meaningless. Every series must first be transformed into an economically interpretable quantity and then normalized to a comparable scale.

4.3 Core statistical idea

z_t = (x_t - mean_t) / std_t

A z-score says how far the current observation is from its benchmark in standard-deviation units.

A reading of 0 means "near the estimated benchmark," not "good." A reading of +1 means roughly one historical standard deviation above the benchmark. A reading of -1 means roughly one historical standard deviation below it.

4.4 Robust alternatives

Macro series often contain crises, fat tails, structural changes and non-normal distributions. CHronoZ therefore should not blindly use one z-score formula for everything.

Method	Useful when	Main caution
Mean + standard deviation	Distribution is reasonably stable and symmetric	Sensitive to extreme crises/outliers
Median + MAD	Outliers or fat tails distort the mean	Still assumes historical sample is relevant
Percentile / empirical CDF	Distribution shape is irregular	Less intuitive for linear aggregation
Rolling / expanding benchmark	Economic structure evolves through time	Window choice can move the meaning of neutral
Explicit economic anchor	Series has a real reference point, e.g. PMI 50 or policy target	Anchor may change across regimes

robust_z_t = (x_t - median_t) / (1.4826 * MAD_t)

A robust standardization candidate when extreme observations would otherwise dominate the benchmark.

4.5 Neutral is variable-specific

Neutral method	Example interpretation
Historical center	Typical transformed value over the valid historical sample
Economic anchor	PMI = 50 separates expansion from contraction by construction

Neutral method	Example interpretation
Policy/reference anchor	Inflation gap relative to a documented target, when the target is relevant
Trend-relative	Deviation from a long-run trend when the raw level naturally grows over time

Do not confuse neutral with healthy
A neutral benchmark describes historical reference, not moral quality. For example, historically typical credit growth can still be undesirable if leverage is already excessive.

5. Scope, Variables & Bias Control

A benchmark only becomes reliable when its scope is explicit. CHronoZ should begin country by country, not by mixing unrelated economies into one average. The initial quantitative build should use one country - preferably the United States because of long histories, release metadata, market depth and real-time vintage availability - and then generalize.

5.1 Country-first architecture

State_{country,t} = [Growth, Inflation, Labor, Credit, Policy/Liquidity, External, Structural, ...]

Global aggregation comes later and must use an explicit weighting rule.

A future Global State could use GDP-PPP weights, trade weights, market-cap weights or asset-specific exposure weights. The correct choice depends on the research question; there is no universal global weight.

5.2 Initial macro nodes - provisional, not locked

Node	What it tries to measure	Candidate sub-blocks
Growth	Current real activity relative to trend	Production, consumption, income, business activity, housing
Labor	Demand and slack in the labor market	Payrolls, unemployment, claims, hours, openings
Inflation	Current price pressure and persistence	Core, goods, services, housing, wages
Credit	Private financing availability and balance-sheet stress	Loan growth, lending standards, demand, delinquencies, refinancing
Policy / Liquidity	Policy stance and liquidity provision	Policy rate, real policy rate, reserves, central-bank balance sheet
External	External demand and cross-border pressure	Trade, export demand, current account, global activity
Structural / Productivity	Slow-moving supply capacity	Productivity, labor supply, investment, demographics

5.3 Reliability: more data is not automatically better

A model can become biased simply because one part of the economy has many more available series. Six inflation measures should not automatically receive six times the influence of one production measure.

- Use hierarchical aggregation: indicator -> sub-block -> macro node.
- Avoid duplicate variables that measure almost the same thing unless their redundancy is intentional.
- Track pairwise correlation so one theme is not counted repeatedly.
- Require minimum data coverage before a node score is emitted.
- Test the model across multiple subperiods, including crises and calmer expansions.
- Document every transformation and sign orientation.
- Do not optimize weights solely to maximize historical trading Sharpe. That would contaminate the measurement layer with a downstream objective.

5.4 Weighting strategy

Node_t = sum_i (w_i * standardized_indicator_{i,t}), with sum_i w_i = 1

V0 can begin with transparent equal weights inside carefully defined sub-blocks; later weighting methods must earn their place out-of-sample.

6. Transforming & Encoding Economic Variables

The benchmark should not be built on raw levels simply because those levels are available. A transformation should answer: "What economic quantity are we actually trying to measure?"

Raw series type	Typical transformation	Why
Nominal or trending level	YoY growth, QoQ annualized growth, log difference	Removes mechanical upward trend
Price index	Inflation rate / short-run annualized inflation	Converts index level into price-pressure measure
Quantity index	Growth rate or deviation from trend	Measures activity rather than absolute scale
Interest rate	Level, real rate, spread or change	Rates already have natural units
PMI / diffusion survey	Distance from 50 and/or standardized history	50 has economic meaning
Unemployment	Level normalized to history; sign can be oriented	Higher unemployment usually implies weaker labor state
Credit	Real/nominal growth, standards, delinquency gap	Separates quantity, access and stress
Spread	Raw spread then standardized	Spread level itself carries information

6.1 Example: year-over-year transformation

$$\text{YoY}_t = 100 * (x_t / x_{(t-12)} - 1)$$

Useful for many monthly level series, but not automatically the best choice for every indicator.

6.2 Direction orientation

CHronoZ should orient transformed indicators so the sign has a consistent meaning inside each node. For Growth, higher industrial production is positive while higher jobless claims may be negative. The raw series is not changed; a documented orientation multiplier is applied only to the standardized state.

$$\text{oriented_z}_t = \text{sign}_i * z_t, \quad \text{where sign}_i \text{ is } +1 \text{ or } -1$$

The sign is metadata and must be auditable.

6.3 Structural breaks

A single historical mean can become misleading after major policy, demographic or institutional changes. CHronoZ should therefore monitor whether the distribution itself changed. Possible tools include rolling statistics, change-point tests, subperiod diagnostics and explicit model-version changes. A structural break should not be hidden by silently changing the baseline.

7. Economic State Engine

After indicators have been transformed and normalized, CHronoZ can build the current state. The preferred object is a vector of economic dimensions, not one total score.

$$S_t = [\text{Growth}_t, \text{Inflation}_t, \text{Labor}_t, \text{Credit}_t, \text{Liquidity}_t, \dots]$$

Each component is measured relative to its own benchmark.

7.1 Why one total economic score is dangerous

If Growth = -1 and Inflation = +1, a simple average equals zero. That does not mean the economy is neutral; it may describe a stagflationary combination. The vector preserves this information.

7.2 Each node should carry more than level

$$\text{Node}_t = [\text{Level}_t, \text{Momentum}_t, \text{Breadth}_t, \text{Coverage}_t]$$

The same level can mean very different things depending on direction and participation.

Field	Plain meaning	Example
Level	Where the node is relative to its benchmark	Growth = -0.4
Momentum	Whether the node is improving or deteriorating	Growth momentum = -0.7
Breadth	How many underlying indicators agree with the move	30% improving
Coverage	How much of the intended node has current data	85% available

7.3 Breadth

$$\text{Breadth}_t = \text{improving_indicators}_t / \text{available_indicators}_t$$

A narrow move and a broad move should not be treated as identical.

7.4 Current state example

Node	Level	Momentum	Breadth	Interpretation
Growth	-0.4	-0.7	30%	Below benchmark and deteriorating broadly
Inflation	+0.8	-0.3	40%	Still above benchmark but cooling
Labor	+0.2	-0.5	35%	Still firm, losing momentum
Credit	-1.2	-0.2	25%	Weak and still deteriorating
Liquidity	+0.5	+0.3	70%	Above benchmark and improving

8. Baseline, Rolling Mean & Momentum

The founder sketch contains two time concepts that should be separated: a slow benchmark that answers "what is normal?" and a faster momentum measure that answers "where are conditions moving?" If the benchmark moves too quickly, a boom or crisis can become "normal" simply because it lasted a few months.

8.1 Rolling benchmark

$\text{mean}_t(W)$ = average of the previous W observations only

During backtests, the current and future observations must not leak into the historical benchmark.

$\text{std}_t(W)$ = standard deviation of the previous W observations

Together, rolling center and dispersion create a time-aware benchmark.

$\text{state}_z_t = (\text{current}_t - \text{mean}_t(W)) / \text{std}_t(W)$

This is the spread between the observed state and the rolling baseline in standardized units.

8.2 Long baseline versus short momentum

Object	Purpose	Typical horizon idea
Long / expanding benchmark	Estimate normal reference distribution	Many years or all valid prior history
Medium rolling benchmark	Adapt cautiously to structural change	Multi-year window
Short momentum	Detect recent direction and acceleration	Several months / releases
Release surprise	Measure new information versus consensus	Single release relative to historical surprise distribution

8.3 Momentum methods to test

- Simple change: $x_t - x_{(t-1)}$.
- Change in change: approximate acceleration in discrete data.
- Slope of the last k observations.
- Short moving average minus long moving average.
- Fast EMA minus slow EMA.
- Momentum of the normalized node rather than momentum of every raw series.

No single momentum formula should be assumed correct. The correct choice is the one that is economically interpretable, stable and useful out-of-sample.

8.4 Founder sketch - revised interpretation

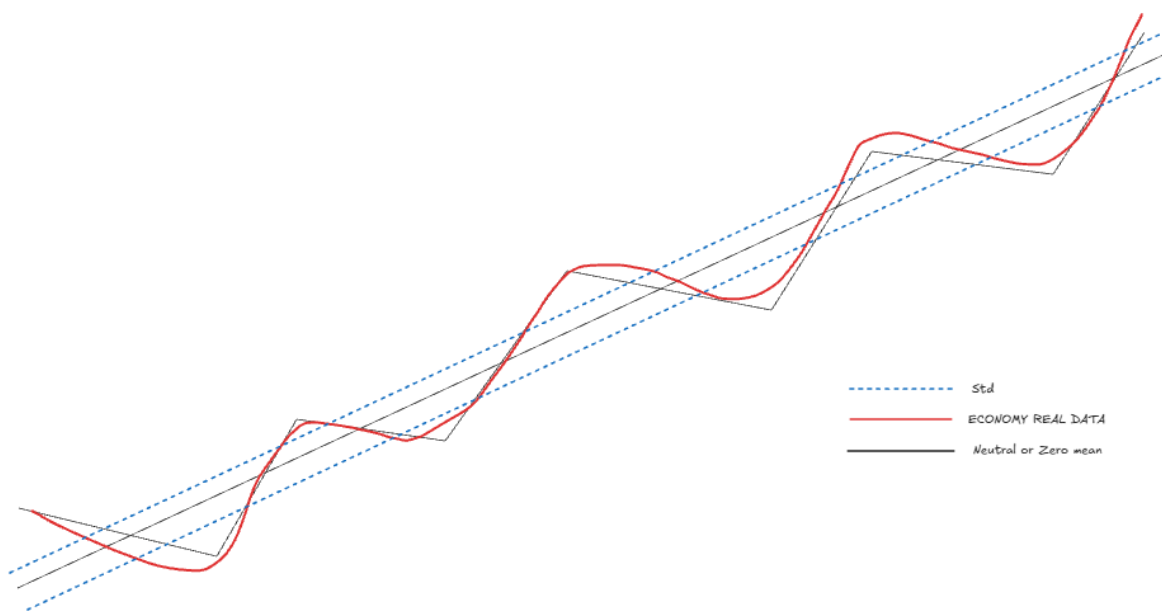


Figure 1. Founder sketch. Recommended reading: red = observed economic state, black = estimated benchmark center, blue dashed = historical dispersion bands. The gap between red and black is the state spread; its change through time is momentum.

9. Regime Discovery & Machine Learning

Machine learning enters only after the state representation is good enough to trust. It should discover patterns in a carefully designed economic state, not rescue a poor measurement system.

9.1 Regime model input

```
R_t = [ Growth_level, Growth_momentum, Inflation_level, Inflation_momentum,  
        Credit_level, Credit_momentum, ... ]
```

Regime discovery should normally use standardized state features rather than raw CPI/GDP levels.

9.2 Recommended sequence

1. Visualize state history and inspect distributions.
2. Use PCA as an exploratory compression tool, not as economic truth.
3. Test nearest-neighbor / historical-analogue retrieval.
4. Explore clustering such as K-Means for simple structure.
5. Use Gaussian Mixture Models when regimes overlap probabilistically.
6. Use Hidden Markov Models only when temporal transitions add value.
7. Validate stability out-of-sample before naming or operationalizing a regime.

9.3 Labels come after structure

A cluster should initially be Cluster 0, Cluster 1, Cluster 2. Only after examining its typical Growth, Inflation, Credit and Liquidity characteristics should a human or LLM give it an economic label such as "disinflationary slowdown." The label is interpretation; the cluster is the statistical object.

Anti-overfitting rule

If a regime exists only because a particular clustering seed or sample window creates it, it is not yet a reliable regime. Stability matters more than a pretty historical chart.

10. Comparing Dump Evidence with the Economic Model

Structured text evidence is an overlay on the measured state. A sentence from a report should not directly set Credit = -1.2 or Growth = +0.4. Instead, CHronoZ maps the finding to the relevant node, checks current measurements, identifies missing information and then forms a research hypothesis.

10.1 Evidence relation labels

Relation	Meaning
Supports	Evidence is directionally consistent with the measured state or existing thesis
Contradicts	Evidence points against the measured state/thesis and deserves investigation
Adds new dimension	Evidence covers something not captured well by current quantitative series
Already known / redundant	Evidence repeats information already present in state or prior research
Potentially relevant	Connection exists but evidence is too weak or indirect
Irrelevant	No credible link to the current research question

10.2 Example: credit

Measured state
Credit level = -1.2
Credit momentum = -0.2
Breadth = 25% improving

New dump finding
"Bank lending standards tightened."

CHronoZ should not stop at "credit bearish." It should investigate loan demand, corporate bond issuance, refinancing conditions, defaults, bank balance sheets, capex and possible fiscal offsets. The correct conclusion may be that bank credit is tightening while bond financing partly offsets the shock.

10.3 Evidence as Bayesian-like updating - without fake precision

Conceptually, new evidence updates a prior worldview. But until CHronoZ has enough historical research instances to estimate real likelihoods, it should avoid arbitrary numeric belief updates such as "+0.23 confidence." V0 should use transparent qualitative relations and store the eventual outcome for future calibration.

11. Causality as a Research Hypothesis

v0.1 used a transmission graph to constrain reasoning. v0.2 keeps the useful idea of known mechanisms but changes its status: the map is a research library, not a deterministic graph of economic truth.

11.1 Candidate mechanism library

CHronoZ can store common mechanisms such as policy rates -> borrowing costs -> credit demand -> spending. These links help the agent know what to investigate. They do not mean the effect must occur, must have a fixed sign, or must have a fixed lag in every state.

11.2 Causal hypothesis object

- Claim: what causal relationship is being proposed?
- Mechanism: why could A affect B?
- Temporal fit: did the proposed cause happen before the effect?
- Supporting evidence: what independent observations fit the mechanism?
- Contradictory evidence: what important facts do not fit?
- Alternative explanations: what else could produce the same pattern?
- State dependence: why might the mechanism be stronger or weaker now?
- Observable falsifier: what future observation would weaken or reject the thesis?

11.3 State-dependent causality

Impact = f(event, starting state, transmission strength, lag, offsets)

A 25bp rate shock is not economically identical in a low-leverage economy and a highly leveraged refinancing-sensitive economy.

Rule

Correlation can help discover relationships. It cannot, by itself, be promoted to causal truth. CHronoZ should preserve the gap between statistical association and economic mechanism.

12. Expectations & Market Pricing Model

Economic reality is not the same as a tradable opportunity. A weak economy can still produce a bad bond trade if the bond market already prices an even worse economy. CHronoZ therefore separates three objects: what is happening, what was expected, and what is already priced.

REALITY vs CONSENSUS vs MARKET PRICING

Trade direction should come from the gap, not from the economic state alone.

12.1 Release surprise

$$\text{surprise}_t = \text{actual}_t - \text{consensus}_t$$

A raw surprise can be standardized by its historical surprise distribution.

$$\text{surprise}_z_t = (\text{actual}_t - \text{consensus}_t) / \text{std}(\text{historical surprises})$$

This makes surprises across releases more comparable.

12.2 Market Pricing State - separate from Economic State

Pricing block	Candidate measures
Rates	Market-implied policy path, curve slope, real yields, term structure
Inflation pricing	Breakevens, inflation swaps, real-vs-nominal rate structure
Credit pricing	IG/HY spreads, CDS, default compensation, issuance conditions
Equity pricing	Valuation, earnings revisions, equity risk measures, volatility
FX pricing	Rate differentials, carry, relative growth expectations, dollar conditions
Commodities	Spot/futures curve, inventory/supply-sensitive pricing
Volatility / positioning	Implied volatility, skew, positioning where available

Keeping market prices outside the real-economy benchmark reduces circularity. CHronoZ can then ask whether market pricing confirms, contradicts or overreacts to the measured state.

12.3 Pricing gap is asset-specific

There should not be one universal "fair value gap" formula for every asset. Rates require a policy-path comparison; inflation markets require an inflation-distribution comparison; credit requires macro/default risk versus spreads; equities require growth, earnings, liquidity and valuation. The common idea is consistent even when the formulas differ: compare the research-implied distribution with the market-implied distribution.

13. Cross-Asset Interpretation

Cross-asset analysis asks whether different markets are telling the same story. Rates, FX, credit, equities, commodities, gold and volatility may confirm one another or reveal an internal contradiction.

13.1 Measurement first, interpretation second

The quantitative engine should first standardize market moves relative to their own history. The LLM then interprets the configuration. "Gold is confirming" should not be a free-form phrase; it should refer to a measurable move and a clearly stated thesis.

13.2 Example configuration

Economic / market object	Observed direction	Possible reading
Growth momentum	Down	Real economy losing speed
Inflation momentum	Down	Disinflation continuing
2Y yield	Down strongly	Market prices easier policy
USD	Down	Easier relative policy / weaker dollar impulse
Gold	Up	Consistent with lower real-rate / defensive demand story
HY spreads	Flat	Credit market not pricing severe recession
Equities	Up	Risk assets still discount benign slowdown / liquidity support

The contradiction between weak growth and resilient credit/equity pricing may be more useful than a single directional signal. CHronoZ should surface the disagreement rather than force all assets into one narrative.

14. Investment & Trade Decision Layer

The decision layer is downstream of measurement and research. Its purpose is not to prove that the economic benchmark is "correct" by finding a profitable backtest. Instead, it asks whether the state, momentum, surprise and pricing gap contain useful information for future asset returns.

14.1 Research state versus trade signal

Research object	Decision translation
Economic State	What conditions are currently true relative to benchmark?
Momentum	What is changing and how broad is the change?
Surprise	What new information differed from consensus?
Pricing State	What does the market already appear to expect?
Evidence overlay	Does new qualitative information support, contradict or expand the thesis?
Decision rule	Is the expected reward large enough relative to uncertainty, transaction cost and risk?

14.2 A simple empirical mapping

$$\text{future_return_}(\text{asset}, t+h) = \alpha + \beta * \text{State}_t + \gamma * \text{Momentum}_t + \delta * \text{Pricing}_t + \text{error}$$

This is an example research equation, not a universal production model.

Later models may be nonlinear or regime-dependent. But the first versions should remain interpretable enough that we know what is producing the signal.

14.3 Applied use cases

- Rates: compare research-implied policy path with the market-implied path.
- Gold: study inflation, real rates, liquidity, dollar conditions and risk regime jointly.
- Equities: study growth, earnings revisions, liquidity, credit and valuation rather than "GDP up = stocks up."
- FX: focus on relative states between economies, rate differentials and market pricing.
- Credit: compare measured default/refinancing risk with spread compensation.
- Multi-asset allocation: use regime/state as a conditioning variable rather than a single buy/sell switch.

15. Backtesting, Data Vintages & Calibration

A macro model is unusually vulnerable to hindsight. Final revised GDP, unemployment or inflation data can be materially different from what investors actually knew at the time. A credible CHronoZ backtest must reproduce the historical information set as closely as practical.

15.1 Walk-forward rule

InformationSet(t) = only information observable at time t

The benchmark at t must be trained only on history available before t.

- 1. Choose historical date t.
- 2. Load the data vintage that would have been available at t.
- 3. Fit/refresh the benchmark using only observations available before t.
- 4. Calculate state, momentum, breadth and regime at t.
- 5. Load consensus and market pricing known at t.
- 6. Generate the research or trade signal.
- 7. Execute only after the relevant release/observation became available.
- 8. Observe future economic outcomes and asset returns.
- 9. Advance to the next date and repeat.

15.2 Vintage-aware data

Real-time historical databases such as ALFRED are important because they preserve data as it existed on historical vintage dates, including later revisions. This prevents revised-data leakage from making the historical model look smarter than it could have been in real time.

15.3 Two levels of validation

Economic validation	Investment validation
Does the state behave sensibly across known cycles?	Does the state/pricing gap contain information about future returns?
Are transforms stable across subperiods?	Does performance survive transaction costs?
Does the benchmark remain interpretable after crises?	Does it survive out-of-sample and different assets?
Are turning points / momentum changes sensible?	Sharpe, Sortino, max drawdown, hit rate, turnover, beta/alpha
Are revisions and missing data handled correctly?	Does the result depend on one regime or one lucky period?

15.4 Ablation test: does the text layer add value?

Model	Information included
A	Economic State only
B	State + Momentum/Breadth
C	+ Expectations & Market Pricing
D	+ Structured Dump Evidence
E	+ Causal Research / Critic

If Model D barely improves out-of-sample performance over Model C, the correct conclusion is not to force a story. It means the current text-evidence layer adds little incremental predictive value. That is a research result. If it materially improves performance, CHronoZ can then investigate which evidence classes created the improvement.

16. Temporal Learning & Internal Economic R&D Dataset

The durable advantage of CHronoZ is not the prompt. It is the historical research dataset created every time the system forms a view and later observes what happened.

State(t) + Evidence(t) + Outlook(t) + Pricing(t) -> Future Outcome(t+h) -> Error / Calibration

Every timestamp becomes a training and evaluation record.

16.1 Store at every research timestamp

- Economic state vector and node diagnostics.
- Exact data vintage / as-of timestamp when practical.
- Information dumps and extracted findings.
- Causal hypotheses, alternatives and falsifiers.
- Consensus expectations and market-pricing state.
- Macro outlook and thesis version.
- Any investment/trade signal and intended horizon.
- Observed future economic outcomes and asset returns.
- Error, calibration result and post-mortem notes.

16.2 What this enables later

- Historical analogue retrieval: "When has the state looked like this before?"
- Regime-transition research and transition probabilities.
- Evidence-type calibration: which kinds of research dumps actually matter?
- Forecast calibration and Brier/log-loss evaluation for probabilistic outlooks.
- Causal-hypothesis post-mortems: which mechanisms repeatedly worked or failed?
- Model-version comparison through time.

17. What Uses LLM, What Uses Algorithms?

CHronoZ is deliberately hybrid. The simplest rule is: use algorithms where the answer must be reproducible and numeric; use LLMs where the task requires language understanding, research planning, hypothesis generation or synthesis.

Stage	LLM / agent	Algorithm / statistics	Primary owner
Dump-to-Text	Read semantics, charts, claims	Parsing, schema validation	Hybrid
Evidence extraction	Yes	Validation / deduplication	LLM-heavy
Macro classification	Yes	Ontology constraints	LLM-heavy
Transforms / normalization	No	Yes	Python
Benchmark	No	Yes	Python/statistics
State / momentum / breadth	No	Yes	Python
Regime discovery	Interpret labels only	PCA/GMM/HMM/NN etc.	Python/ML
Causal research	Yes	Statistical tests may support	LLM-heavy
Expectations	Interpret	Actual-consensus calculations	Hybrid
Market pricing	Interpret	Curves/spreads/state calculations	Hybrid
Cross-asset	Explain confirmation/divergence	Standardize asset moves	Hybrid
Thesis / critic	Yes	Model outputs as evidence	LLM
Journal	Yes	No new reasoning	LLM
Backtest / calibration	No	Yes	Python
Memory / versioning	Optional semantic tags	Storage/index/version control	Infrastructure

Production rule
LLM may call the quantitative engine. It should not reimplement benchmark math inside prose. The quantitative engine may provide evidence to the LLM. It should not write the causal story for the LLM.

18. Python/Jupyter Development & Agent Integration

The quantitative engine should be developed independently from the Wonnyy/TypeScript application layer. Jupyter is the laboratory; Python modules are the reusable model; JSON/API is the contract to CHronoZ agents.

18.1 Recommended quantitative repository

```
chronoz-quant/  
notebooks/{01_data_exploration, 02_normalization, 03_state_vector, 04_regime_exploration}.ipynb  
src/chronoz_quant/{data, transforms, benchmark, state, nodes, regime, pricing, backtest, calibration}/  
tests/  
data/
```

18.2 Notebook versus source code

Notebook	src/*.py
Explore distributions	Reusable functions
Plot and question assumptions	Calibrated transformations
Try alternative windows/metrics	Benchmark/state engine
Inspect failures visually	Tests and deterministic contracts
Learn the math while coding	Production logic called by other systems

18.3 Integration stages

- 1. V0 - local script: TypeScript runs a Python process and reads JSON output.
- 2. V1 - local API: Python/FastAPI exposes endpoints such as /state, /regime and /pricing.
- 3. V2 - independent quant service: Wonnyy, CHronoZ, dashboards or portfolio systems can consume the same economic engine.

18.4 Example output contract

Field	Example
as_of / economy	2026-09-16 / US
model_version	benchmark-v0.2
Growth	level -0.42 momentum -0.18 breadth 0.30
Inflation	level +0.76 momentum -0.31 breadth 0.40
Credit	level -1.08 momentum -0.12 breadth 0.25
metadata	benchmark: expanding vintage: 2026-09-16

19. One CHronoZ Run - v0.2

1. Receive local research dump.
2. Dump-to-Text / multimodal extraction converts sources into canonical text and provenance.
3. Extract and classify Evidenceltem / Finding objects.
4. Load the latest quantitative Economic State, benchmark metadata and market-pricing state.
5. Map each finding to candidate nodes without directly changing numeric state.
6. Classify the finding as supporting, contradicting, new, redundant, potentially relevant or irrelevant.
7. Identify information gaps and optionally pull external data/research.
8. Generate causal hypotheses, alternatives, contradictions and falsifiers.
9. Compare actual economic conditions with consensus expectations and market pricing.
10. Run cross-asset interpretation using measured market moves.
11. Form/update the Macro Outlook and current thesis.
12. Run Critic / Falsification.
13. If required, translate research into a separate investment/trade decision object.
14. Synthesize the Markdown Macro Journal and compact machine state.
15. Librarian stores immutable journal, state, provenance and model version.
16. When future outcomes arrive, update evaluation/calibration records - not the historical journal.
17. Use the accumulated dataset in future analogue, regime and calibration research.

20. Implementation Roadmap

The build should follow the learning order: one real series, then one node, then the state vector, then regime/prediction. Do not begin with HMMs or a large autonomous agent graph.

Milestone	Goal
Q0 - Environment	Python + Jupyter in VS Code; reproducible project structure
Q1 - One Series	Load one real macro series; inspect raw values and release structure
Q2 - Transform	Choose YoY/MoM/level/spread transformation with economic reasoning
Q3 - Benchmark	Mean/std, median/MAD, percentile and expanding/rolling alternatives
Q4 - State	Current z/percentile + direction + momentum
Q5 - One Node	Aggregate several indicators into one interpretable node
Q6 - Multi-Node State	Growth, Inflation, Labor, Credit etc. as a state vector
Q7 - Vintage Backtest	Walk-forward historical state using real-time vintages where possible
Q8 - Regime Exploration	Nearest neighbors, PCA, clustering, GMM; HMM later if justified
Q9 - Pricing Engine	Rates/inflation/credit/equity/FX pricing states
Q10 - Decision Tests	Asset-return mapping, costs, out-of-sample validation
Q11 - Dump Overlay	Structured evidence compared to state; causal research loop
Q12 - Ablation	Measure incremental value from pricing, text and causal research
Q13 - Deployment	Stable JSON/API contract into CHronoZ TypeScript agent layer

21. Epistemic Discipline & Failure Modes

Failure mode	CHronoZ rule
Look-ahead bias	Never train or normalize with future observations in a historical test
Revised-data leakage	Use vintage/as-of data when available; record vintage metadata
Raw-series nonsense	Transform variables before normalization; do not average incompatible units
Outlier-distorted neutral	Compare mean/std with robust alternatives and inspect distributions
Structural-break blindness	Test rolling/subperiod stability; version changes explicitly
Duplicate-variable bias	Use hierarchy and correlation checks so one theme is not counted repeatedly
One-score oversimplification	Keep the state vector and node diagnostics visible
Hard-coded causal truth	Treat transmission links as candidate mechanisms; require research evidence
Correlation = causation	Keep statistical association separate from causal mechanism
Market circularity	Separate real-economy state from market-pricing state
Overfit regimes	Demand out-of-sample stability and sensible transitions
Fake confidence	Do not emit precise probabilities without calibration
Narrative override	LLM cannot overwrite numeric state because a story sounds persuasive
Backtest over-optimization	Economic benchmark validity comes before trading Sharpe
Text-evidence vanity	Use ablation tests; if dump evidence adds no value, record that result

Insight quality test

A CHronoZ output should not be called an insight unless it can answer several of these questions: What changed? Relative to what benchmark? How broad is the move? Is momentum improving or deteriorating? What was expected? What is priced? What evidence supports or contradicts the view? What mechanism might explain it? What would falsify it? What happened next?

Candidate US Indicator Universe - Research Starting Point

This is a deliberately small starting universe, not a locked ontology. Each series must still pass a data-quality, transformation and redundancy review before production use.

Node	Candidate indicator	Possible encoding
Growth	Industrial Production	MoM / YoY growth; standardized
Growth	Real Personal Consumption Expenditures	YoY / short-run annualized growth
Growth	Real Personal Income	YoY growth
Growth	Real Manufacturing & Trade Sales	YoY growth
Growth	Housing Starts / Permits	YoY or standardized level/change
Growth	ISM / PMI	Deviation from 50 + historical standardization
Labor	Nonfarm Payrolls	Monthly change scaled to history / working-age population
Labor	Unemployment Rate	Standardized level; negative orientation for labor strength
Labor	Initial Jobless Claims	Standardized level/change; negative orientation
Labor	Average Weekly Hours	Change / standardized level
Labor	Job Openings	Level relative to labor force or unemployed
Inflation	Core PCE	YoY + 3m/6m annualized variants
Inflation	Core CPI / services measures	YoY + short-run annualized
Inflation	Shelter inflation	YoY / momentum
Inflation	Wage growth	YoY / short-run momentum
Credit	Commercial & Industrial loan growth	YoY growth
Credit	Consumer credit growth	YoY / real growth
Credit	SLOOS lending standards	Survey balance standardized
Credit	Loan demand	Survey balance standardized
Credit	Delinquency/default measures	Standardized level and momentum
Policy/Liquidity	Policy rate	Level / change / real-policy gap
Policy/Liquidity	Bank reserves	Growth / standardized level
Policy/Liquidity	Central-bank balance sheet	Growth / trend-relative
External	Real exports / imports	Growth and relative demand
Structural	Labor productivity	YoY / multi-quarter trend

Formula & Concept Glossary

Object	Formula / representation	Plain meaning
YoY growth	$100 * (x_t / x_{(t-12)} - 1)$	Change versus one year earlier
Simple change	$\Delta x_t = x_t - x_{(t-1)}$	Latest direction
Z-score	$(x_t - \text{mean}_t) / \text{std}_t$	Distance from benchmark in standard-deviation units
Robust z	$(x_t - \text{median}_t) / (1.4826 * \text{MAD}_t)$	Distance using robust center/dispersion
Node score	$\text{sum}(w_i * z_{i,t})$	Combine comparable standardized indicators
Breadth	improving / available	How many components agree with the move
State vector	[Growth, Inflation, Labor, Credit, ...]	Multidimensional economy at time t
Surprise	actual - consensus	New information relative to expectation
Surprise z	$\text{surprise} / \text{historical std}(\text{surprise})$	Comparable surprise magnitude
Future-return test	$r_{(t+h)} = f(\text{state, momentum, pricing})$	Does current research contain forward information?

Minimum Quantitative Output Contract

The contract should expose enough metadata that an LLM or application can use the output without reconstructing the calculation from prose.

- as_of timestamp and economy
- model version and benchmark method
- data vintage / latest observation date
- indicator-level transformed value, normalized value and percentile where relevant
- node-level level, momentum, breadth and coverage
- regime probabilities / nearest analogues when available
- warnings for stale or missing data
- source/provenance identifiers
- no hidden LLM-generated numeric overrides

Recommended machine objects

Object	Purpose
BenchmarkState	Center, dispersion, distribution metadata per indicator/node
EconomicState	Current real-economy state vector
MomentumState	Recent direction/acceleration diagnostics
RegimeState	Statistical regime probabilities / analogues
PricingState	Market-implied expectations and cross-asset measurements
EvidenceOverlay	Structured findings and relation to current thesis/state
ResearchOutlook	LLM-generated hypothesis/thesis with evidence and falsifiers
DecisionState	Optional trade/investment expression with horizon and risk
EvaluationRecord	Observed future outcome and model error

Methodological References & Inspiration

These references are methodological precedents, not templates that CHronoZ must copy. The final model should remain transparent and should be validated against its own objective.

[1] Federal Reserve Bank of Chicago - Chicago Fed National Activity Index (CFNAI): About the CFNAI

A real example of combining many standardized activity indicators into an index with mean zero, standard deviation one, a moving-average view and a diffusion/breadth concept.

<https://www.chicagofed.org/research/data/cfnai/about>

[2] Federal Reserve Bank of Chicago - CFNAI methodology / principal-components discussions

Illustrates factor/PCA weighting as one possible way to summarize common variation across indicators.

<https://www.chicagofed.org/publications/chicago-fed-letter/2013/june-311>

[3] ALFRED - Archival Federal Reserve Economic Data

Supports vintage-date retrieval so historical backtests can use data as it actually existed at a past date.

<https://alfred.stlouisfed.org/help/downloaddata>

[4] OECD - Composite Leading Indicators FAQ

Example of explicit country aggregation and weighting methodology, including GDP-PPP weights for zone aggregates.

<https://www.oecd.org/en/data/insights/data-explainers/2024/04/composite-leading-indicators-frequently-asked-questions.html>

Final principle

Measure first. Interpret second. Price third. Decide fourth. Then wait for reality and score the error.

CHronoZ v0.2 - A measurable way to remember how the world changes.